| | | | | |
|---|---|---|---|---|
| **5** | **(a) (i)** | in series $2X$ <u>or</u> in parallel $X/2$ | M1 | |
| | | other relationship given <u>and</u> $4\times$ greater in series (than in parallel) | A1 | [2] |
| | **(ii)** | due to the internal resistance | B1 | |
| | | total resistance for series circuit is not four times greater than resistance for parallel circuit | B1 | [2] |
| | **(iii)** | **1.** $E = I_1(2X + r)$ or $12 = 1.2(2X + r)$ | A1 | |
| | | **2.** $E = I_2(X/2 + r)$ or $12 = 3.0(X/2 + r)$ | A1 | [2] |
| | **(iv)** | $2X + r = 10$ and $X/2 + r = 4$<br>$X = 4.0\,\Omega$ | A1 | [1] |

| | | | |
|---|---|---|---|
| **(b)** | $P = I^2R$ or $V^2/R$ or $VI$ | C1 | |
| | ratio $= [(1.2)^2 \times 4] / [(1.5)^2 \times 4]$<br>$= 0.64$ | A1 | [2] |
| **(c)** | the resistance (of a lamp) changes with $V$ or $I$ | B1 | |
| | $V$ or $I$ is greater in parallel circuit or circuit 2<br>*or* $V$ or $I$ is less in series circuit or circuit 1 | B1 | [2] |